# Waste Management & Recycling: System Design Description

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| Document number | SHV-WM-SDD-0001 |
| Revision | C |
| Status | APPROVED |
| Classification | INTERNAL |
| Author | Tyce Easton |
| Approved by | Selene Abaroth |
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## 1. System overview

Waste management collects, compacts, treats and recycles station waste through compactor units, treatment plants and a materials recycling line.

The system is led by Life Support & Habitation. It is delivered through 5 work packages with a combined budget of 69,799.76 Mcr.

## 2. Components

| Code | Component | Specification | Drawing | Bill of materials |
|---|---|---|---|---|
| WM-TCU | Compactor unit | SHV-WM-SPEC-0101 | SHV-WM-DWG-0201 | SHV-WM-BOM-0301 |
| WM-WTP | Waste treatment plant | SHV-WM-SPEC-0102 | SHV-WM-DWG-0202 | SHV-WM-BOM-0302 |
| WM-RCY | Materials recycling line | SHV-WM-SPEC-0103 | SHV-WM-DWG-0203 | SHV-WM-BOM-0303 |

## 3. Architecture

### 3.1 Compactor unit

The compactor unit is specified in SHV-WM-SPEC-0101. Its governing parameter is the compaction force, set at 11.5 MN. The responsible engineer is Tyce Easton.

### 3.2 Waste treatment plant

The waste treatment plant is specified in SHV-WM-SPEC-0102. Its governing parameter is the throughput, set at 1,951 t/day. The responsible engineer is Treva Bresk.

### 3.3 Materials recycling line

The materials recycling line is specified in SHV-WM-SPEC-0103. Its governing parameter is the recovery rate, set at 80.9 %. The responsible engineer is Ilka Warrick.

## 4. Interfaces

| Document | Title |
|---|---|
| SHV-ICD-LS-WM | Interface Control Document: Life Support & Atmospherics / Waste Management & Recycling |

## 5. Work packages

| Work package | Name | Owner | Status | Complete |
|---|---|---|---|---|
| WP-SHV-WM-01 | Waste Management & Recycling: Design & analysis | Noa Venner | IN_PROGRESS | 82.3% |
| WP-SHV-WM-02 | Waste Management & Recycling: Fabricate & install compactor unit | Gessa Fiore | IN_PROGRESS | 72.7% |
| WP-SHV-WM-03 | Waste Management & Recycling: Fabricate & install waste treatment plant | Theron Wolcott | IN_PROGRESS | 78.0% |
| WP-SHV-WM-04 | Waste Management & Recycling: Fabricate & install materials recycling line | Ines Everhart | IN_PROGRESS | 88.0% |
| WP-SHV-WM-05 | Waste Management & Recycling: Integration & test | Callum Yarrow | IN_PROGRESS | 52.7% |

## 6. Risks

- Supply of compactor unit assemblies depends on a single supplier.
- Supply of waste treatment plant assemblies depends on a single supplier.
- Integration slots in the station schedule are shared with neighbouring zones.